

Data modeling: CSCI E-106

Building the Regression Model III: Remedial Measures

Overview

remedial measures may need to be taken:

- a regression model is not appropriate
- several cases are very influential

previous: transformations to linearize the regression relation

- the error distributions more **nearly normal**
- make the variances of the error terms more **nearly equal**

Remedial Measures

remedial measure to deal with:

- unequal error variances
- a high degree of multicollinearity
- influential observations

two methods for nonparametric regression:

- lowess
- regression trees

bootstrapping: for evaluating the precision of the complex estimators

Weighted Least Squares (WLS)

- transformation of Y - reducing or eliminating unequal variances
- **difficulty:** may create an inappropriate regression relationship
- **weighted least squares**
 - when an appropriate regression relationship has been found but the **variances of the error terms are unequal**

Weighted Least Squares (WLS), cont'd

$$Y_i = \beta_0 + \beta_1 X_{i,1} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i, \quad i = 1, \dots, n$$

- parameters: $\beta_0, \beta_1, \dots, \beta_{p-1}$
- known constants: $X_{i,1}, \dots, X_{i,p-1}$
- $\varepsilon_i \sim N(0, \sigma_i^2)$, independently normally distributed

$$\Rightarrow \underset{n \times n}{\sigma^2\{\varepsilon\}} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{bmatrix}$$

- $b = (X'X)^{-1}X'Y \Rightarrow$ unbiased, consistent but don't have the minimum variance

Weighted Least Squares (WLS), cont'd

- Consider: Observations with **small variances** provide **more reliable information** about the regression function than those with large variances.
- Error Variances **Know** ($w_i = \frac{1}{\sigma_i^2}$) $\Rightarrow$ **UNREALISTIC!**
- Error Variances **Know up to Proportionality Constant** ($w_i = k \frac{1}{\sigma_i^2}$)
- Error Variances **Unknown**: estimation of variance function or standard deviation function ($w_i = \frac{1}{(\hat{s}_i)^2}$, $w_i = \frac{1}{\hat{v}_i}$)

WLS – Error Variances Known

Methods of maximum likelihood to obtain estimators:

- $\varepsilon_i \stackrel{\text{indep.}}{\sim} N(0, \sigma_i^2)$ and $w_i = \frac{1}{\sigma_i^2}$

$$\begin{aligned}\Rightarrow L(\beta) &= \prod_{i=1}^n \frac{1}{(2\pi\sigma_i^2)^{1/2}} \exp \left[-\frac{1}{2\sigma_i^2} (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_{p-1} X_{i,p-1})^2 \right] \\ &= \left[\prod_{i=1}^n \left(\frac{w_i}{2\pi} \right)^{1/2} \right] \exp \left[-\frac{1}{2} \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_{p-1} X_{i,p-1})^2 \right]\end{aligned}$$

$$\Rightarrow \mathbf{b} = \arg \max_{\beta} L(\beta)$$

$$\Leftrightarrow \mathbf{b} = \arg \min_{\beta} Q_w \quad (Q_w = \sum_{i=1}^n w_i (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_{p-1} X_{i,p-1})^2)$$

- Called *weighted least squares criterion* (min Q_w)
- $w_i = 1 \Rightarrow$ OLS

WLS – Error Variances Known, cont'd

- $w_i = \frac{1}{\sigma_i^2}$: reflects the amount of information contained in the observation Y_i
- $\text{var } Y_i$ large $\Rightarrow w_i$ less

$$\mathbf{W}_{n \times n} = \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & w_n \end{bmatrix}$$

- The normal equations: $(\mathbf{X}'\mathbf{W}\mathbf{X})\mathbf{b}_w = \mathbf{X}'\mathbf{W}\mathbf{Y}$
- The weighted least squares (MLE) for β :

$$\begin{aligned} \mathbf{b}_w &= (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{Y} \\ \Rightarrow \sigma^2\{\mathbf{b}_w\} &= (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1} \quad (\because \sigma^2\{\mathbf{Y}\} = \mathbf{W}^{-1}) \\ &\quad p \times p \end{aligned}$$

WLS – Error Variances Known, cont'd

- $\mathbf{b}_w$: unbiased, consistent, have minimum variance among unbiased linear estimators
- When the weights are known, $\mathbf{b}_w$ generally exhibits **less variability** than $\mathbf{b}$.

WLS - Error Variances Known Up To Proportionality Constant

$$w_i = k \frac{1}{\sigma_i^2}$$

$$\mathbf{b}_w = (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1} \mathbf{X}' \mathbf{W} \mathbf{Y} \quad (\text{unaffected})$$

$$\Rightarrow \sigma^2_{\mathbf{b}_w} = k (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1}$$

$p \times p$

$$k: \text{ unknown} \Rightarrow s^2_{\mathbf{b}_w} = MSE_w (\mathbf{X}' \mathbf{W} \mathbf{X})^{-1}$$

$p \times p$

$$\text{where } MSE_w = \frac{\sum w_i (Y_i - \hat{Y}_i)^2}{n - p} = \frac{\sum w_i e_i^2}{n - p}$$

- MSE_w : an estimator of the proportionality constant k

WLS - Error Variances Unknown

- One rarely has knowledge of the variances $\sigma_i^2 \Rightarrow$ estimate of the variances

$$\sigma_i^2 = E(\varepsilon_i^2) - (E(\varepsilon_i))^2$$

- Estimator of σ_i^2 : the squared residual e_i^2
- Estimator of $\sigma_i = \left| \sqrt{\sigma_i^2} \right|$: the absolute residuals $|e_i|$

WLS - Error Variances Unknown, cont'd

the estimation process:

1. Fit the regression model by OLS and analyze e_i
2. Estimate the variance function or the standard deviation function by regressing e_i^2 or $|e_i|$ on the appropriate predictor(s).
3. $\hat{s}_i$ for standard deviation function, $\hat{v}_i$ for the variance function: to obtain w_i
4. obtain b_w by using w_i

(Sometimes, iterated for several times to reach stabilize to stabilize the estimated regression coefficients $\Rightarrow$ iteratively reweighted least squares)

WLS - Error Variances Unknown, cont'd

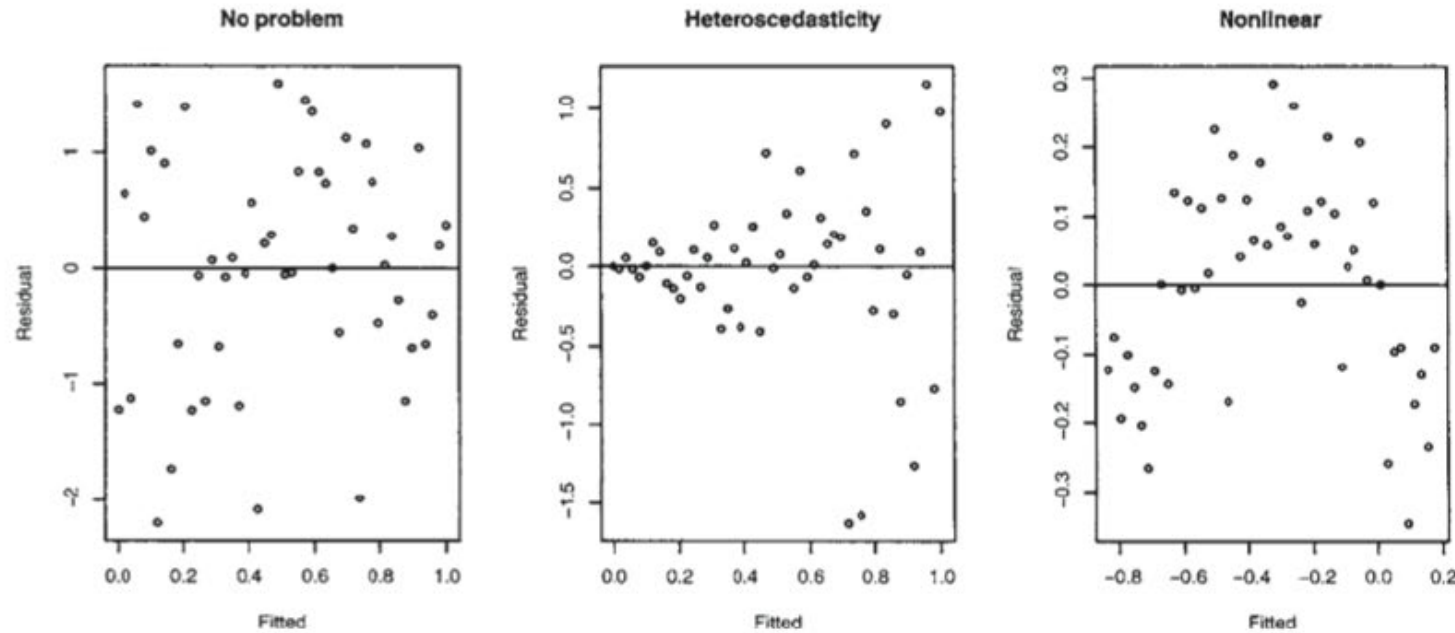


Figure 4.1 *Residuals vs. fitted plots—the first suggests no change to the current model while the second shows nonconstant variance and the third indicates some nonlinearity, which should prompt some change in the structural form of the model.*

- Reference: figure from Faraway's Linear Models with R (2005, p. 59)

WLS - Error Variances Unknown, cont'd

Some possible variance and standard deviation functions:

1. plot e_i vs. X_1 : a megaphone shape $\Rightarrow |e_i|$ regresses on X_1
2. plot e_i vs. $\hat{Y}$: a megaphone shape $\Rightarrow |e_i|$ regresses on $\hat{Y}$
3. plot e_i^2 vs. X_3 : upward tendency $\Rightarrow e_i^2$ regresses on X_3
4. plot e_i vs. X_2 : increase rapidly with X_2 up to a point and then increases more slowly $\Rightarrow |e_i|$ regresses on X_2, X_2^2

$\Rightarrow w_i = \frac{1}{(\hat{s}_i)^2}$: $\hat{s}_i$ is fitted value from standard deviation function

$\Rightarrow w_i = \frac{1}{\hat{v}_i}$: $\hat{v}_i$ is fitted value from variance function

$\Rightarrow$ the weight matrix $W \Rightarrow b_w = (X'WX)^{-1} X'WY$

WLS - Error Variances Unknown, cont'd

Using of Replicated or Near Replicates

- Replicate observations are made at each combination of levels of X s
- If the number of replications (or near replications) is large, w_i may be obtained directly from the sample variances of Y observations at each combination of levels on X variables
- each case in a replicate group receives the same weight with this method
- Confidence interval: (similar but approximate)

$$b_k \pm t(1 - \alpha/2; n - p)s\{b_k\}$$

WLS - Error Variances Unknown, cont'd

Using of Ordinary Least Squares with Unequal Error Variances:

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

$$\Rightarrow \sigma^2\{\mathbf{b}\} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\sigma^2\{\boldsymbol{\varepsilon}\}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$$

$$\Rightarrow \mathbf{S}^2\{\mathbf{b}\} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{S}_0\{\boldsymbol{\varepsilon}\}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$$

$$\text{where } \mathbf{S}_0 = \text{diag}\{e_1^2, e_2^2, \dots, e_n^2\}$$

Blood Pressure Example

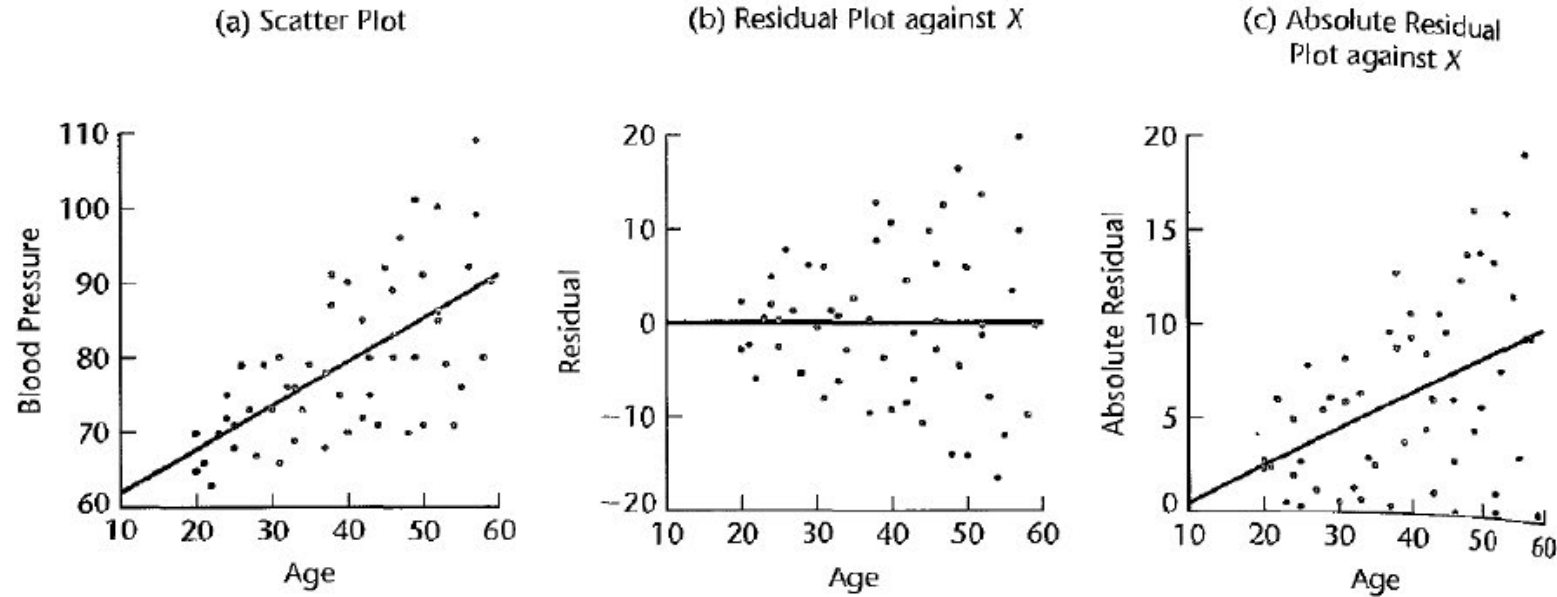
- 54 subjects: 20-60 women

TABLE 11.1
Weighted Least
Squares—
Blood Pressure
Example.

	(1)	(2)	(3)	(4)	(5)	(6)
Subject	Age	Diastolic Blood Pressure				
i	X_i	Y_i	e_i	$ e_i $	$\hat{y}_i$	w_i
1	27	73	1.18	1.18	3.801	.06921
2	21	66	-2.34	2.34	2.612	.14656
3	22	63	-5.92	5.92	2.810	.12662
...	...	...	...	...	...	...
52	52	100	13.68	13.68	8.756	.01304
53	58	80	-9.80	9.80	9.944	.01011
54	57	109	19.78	19.78	9.746	.01053

Blood Pressure Example, cont'd

FIGURE 11.1 Diagnostic Plots Detecting Unequal Error Variances—Blood Pressure Example.



- linear regression function by unweighted least squares:

$$\hat{Y} = \underset{(3.994)}{56.157} + \underset{(0.09695)}{0.58003}X$$

Blood Pressure Example, cont'd

- $|e_i|$ regresses on X : $\hat{s} = -1.54946 + 0.198172X$

$$\hat{s}_1 = 3.801 \Rightarrow w_1 = 1/(\hat{s}_1)^2 = 0.0692$$

- WLS: $\hat{Y} = 55.566 + 0.59634X$
- C.I. for β_1 : $\alpha = 0.05$; $s\{b_{w1}\} = 0.07924$

$$0.59643 \pm \frac{t(0.975; 52)(0.07924)}{=2.007} \Rightarrow 0.437 \leq \beta_1 \leq 0.755$$

Blood Pressure Example, cont'd

1. unequal variances: **heteroscedasticity** and equal variances: **homoscedasticity**
2. R^2 does not have a **clear-cut meaning** for WLS
3. **WLS** may be view as OLS of transformed variables:

$$\begin{aligned} \mathbf{Y} &= \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}\{\boldsymbol{\varepsilon}\} = \mathbf{0}, \sigma^2\{\boldsymbol{\varepsilon}\} = \mathbf{W}^{-1} \\ \Rightarrow \underset{\mathbf{Y}_w}{\mathbf{W}^{1/2}\mathbf{Y}} &= \underset{\mathbf{X}_w\boldsymbol{\beta}}{\mathbf{W}^{1/2}\mathbf{X}\boldsymbol{\beta}} + \underset{\boldsymbol{\varepsilon}_w}{\mathbf{W}^{1/2}\boldsymbol{\varepsilon}} \\ \Rightarrow \mathbf{E}\{\boldsymbol{\varepsilon}_w\} &= \mathbf{0}; \quad \sigma^2\{\boldsymbol{\varepsilon}_w\} = \mathbf{I}; \\ \mathbf{b}_w &= (\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'\mathbf{Y}_w = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{Y} \end{aligned}$$

Blood Pressure Example, cont'd

The weighted least squares normal equations:

$$\begin{aligned}\sum w_i Y_i &= b_{w0} \sum w_i + b_{w1} \sum w_i X_i \\ \sum w_i X_i Y_i &= b_{w0} \sum w_i X_i + b_{w1} \sum w_i X_i^2\end{aligned}$$

The weighted least squares estimators b_{w0} and b_{w1} in (11.9) are:

$$\begin{aligned}b_{w1} &= \frac{\sum w_i X_i Y_i - \frac{\sum w_i X_i \sum w_i Y_i}{\sum w_i}}{\sum w_i X_i^2 - \frac{(\sum w_i X_i)^2}{\sum w_i}} \\ b_{w0} &= \frac{\sum w_i Y_i - b_{w1} \sum w_i X_i}{\sum w_i}\end{aligned}$$

Blood Pressure Example – R Codes

```
g<-lm(Y~X, data=Blood.Pressure)
```

```
par(mfrow=c(2,2))
```

```
plot(g)
```

```
ei<-g$residuals
```

```
abs.ei<-abs(ei)
```

```
g1<-lm(abs.ei~Blood.Pressure$X)
```

```
summary(g1)
```

Call:

```
lm(formula = abs.ei ~ Blood.Pressure$X)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.7639	-2.7882	-0.1587	3.0757	10.0350

Coefficients:

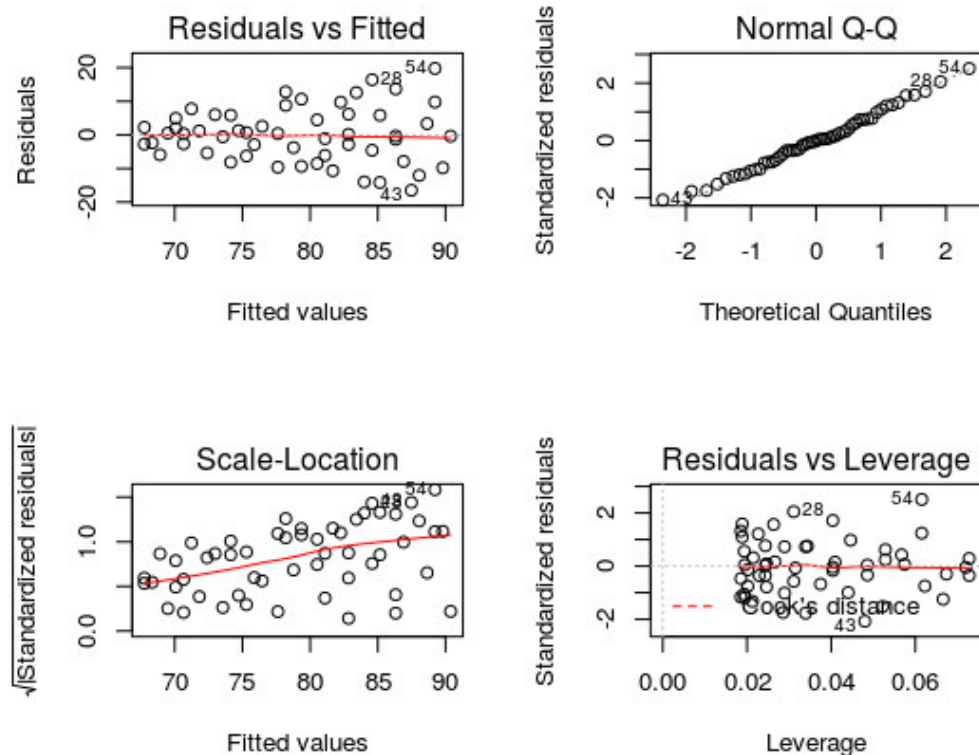
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.54948	2.18692	-0.709	0.48179
Blood.Pressure\$X	0.19817	0.05309	3.733	0.00047 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.461 on 52 degrees of freedom

Multiple R-squared: 0.2113, Adjusted R-squared: 0.1962

F-statistic: 13.93 on 1 and 52 DF, p-value: 0.0004705



Blood Pressure Example – R Codes, cont'd

```
> s<-g1$fitted.values
```

```
> s[1:3]
```

```
      1      2      3
```

```
3.801175 2.612141 2.810313
```

```
> wi=1/(s^2)
```

```
> g2<-lm(Y~X,weights = wi, data=Blood.Pressure)
```

```
> summary(g2)
```

Call:

```
lm(formula = Y ~ X, data = Blood.Pressure, weights = wi)
```

Weighted Residuals:

```
      Min      1Q  Median      3Q      Max
```

```
-2.0230 -0.9939 -0.0327  0.9250  2.2008
```

Coefficients:

```
      Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept) 55.56577    2.52092  22.042 < 2e-16 ***
```

```
X          0.59634    0.07924   7.526 7.19e-10 ***
```

```
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.213 on 52 degrees of freedom

Multiple R-squared: 0.5214, Adjusted R-squared: 0.5122

F-statistic: 56.64 on 1 and 52 DF, p-value: 7.187e-10

Robust Regression

- To examine whether an outlying case is **the result of a recording error**, breakdown of a measurement instrument, or the like.
 - Erroneous data **can be corrected** $\Rightarrow$ corrected
 - Erroneous data **cannot be corrected** $\Rightarrow$ discarded
- Many time: impossible to tell for certain whether the observations for an outlying case are erroneous $\Rightarrow$ not be discarded
- If an outlying influential case is not clearly erroneous $\Rightarrow$ to examine the adequacy of the model

Robust Regression, cont'd

- Numerous robust regression procedures have been developed.
- **LAR or LAD regression**: Least absolute residuals or least absolute deviations

minimum L_1 norm regression

Criterion:

$$\min_{\beta} L_1 = \min_{\beta} \left\{ \sum_{i=1}^n |Y_i - (\beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1})| \right\}$$

- LAR: less emphasis on outlying observations than does the method of least squares
- **Linear programming method**: to obtain the LAR estimators
- LAR method: the residuals will not sum to zero

Robust Regression, cont'd

- LMS regression: Least median of squares

$$\text{median} \left\{ [Y_i - (\beta_0 + \beta_1 X_{i1} + \cdots + \beta_{p-1} X_{i,p-1})]^2 \right\}$$

- IRLS robust regression: Iteratively reweighted least squares method

Iteratively reweighted least squares Regression (IRLS)

IRLS $\Rightarrow$ Weighted Regression

- Outlying cases that have large residuals $\Rightarrow$ given smaller weights.
- The weights are revised as each iteration yields new residuals until the estimation process stabilizes.

A summary of the steps follows:

1. Choose a weight function for weighting the cases.
2. Obtain starting weights for all cases.
3. Use the starting weights in weighted least squares and obtain the residuals from the fitted regression function.
4. Use the residuals in step 3 to obtain revised weights.
5. Continue the iteration until convergence is obtained.

IRLS, cont'd

Weighted Functions:

$$\text{Huber: } w = \begin{cases} 1 & |u| \leq 1.345 \\ \frac{1.345}{|u|} & |u| > 1.345 \end{cases} \quad (11.44)$$

$$\text{Bisquare: } w = \begin{cases} \left[1 - \left(\frac{u}{4.685} \right)^2 \right]^2 & |u| \leq 4.685 \\ 0 & |u| > 4.685 \end{cases} \quad (11.45)$$

$$u_i = \frac{e_i}{MAD}$$

$$MAD = \frac{1}{.6745} \text{ median}\{|e_i - \text{median}\{e_i\}|\}$$

Bootstrapping

- In many nonstandard situations, (ex: nonconstant error variances estimated by iteratively reweighted least squares), standard methods for **evaluating the precision may not be available** or may only be **approximately applicable** when the sample size is large
- Bootstrapping: provide **estimates of the precision of sample estimated** for these complex cases

Bootstrapping, cont'd

general procedure: obtain b_1 by some procedure and wish to evaluate the precision of b_1 by bootstrap method

- The bootstrap method calls for the selection from the observed sample data of a random sample of size n with replacement.
- the bootstrap sample may contain some duplicate data from the original sample and omit some other data in the original sample
- calculates the estimated regression coefficients from the bootstrap sample $\Rightarrow b_1^*$
- repeated a large number of times
- The estimated standard deviation of all of $b_1^* \Rightarrow s\{b_1^*\}$ an estimate of the variability of the sampling distribution of b_1

Bootstrapping, cont'd

Bootstrap Sampling: two basic ways

- When the regression function being fitted is a good model for the data, the error terms have constant variance, and fixed X sampling is appropriate.
- e_i from the original fitting are regarded as the sample data to be sampled with replacement
- After a bootstrap sample with n :

$$\begin{aligned} & e_1^*, \dots, e_n^* \\ \Rightarrow Y_i^* &= \hat{Y}_i + e_i^* \\ Y^* \text{ regression on } X\text{s} &\Rightarrow b_1^* \end{aligned}$$

Bootstrapping, cont'd

- When there is some doubt about the adequacy of the regression function being fitted, the error variance are not constant, and/or *random X sampling* is appropriate.
 - (X_i, Y_i) in the original sample are considered to be the data to be sampled with replacement
 - After a bootstrap sample with n :

n pairs: $(X_1^*, Y_1^*), \dots, (X_n^*, Y_n^*)$

Y^* regression on X^* s $\Rightarrow b_1^*$

Bootstrapping, cont'd

- 200-500 bootstrap samples are adequate
- One can observe the variability of the bootstrap estimates by $s^*\{b_1^*\}$ as the number of bootstrap samples is increased.

Bootstrap Confidence Intervals

- From the bootstrap distribution of b_1^* , find the $\alpha/2$ and $1 - \alpha/2$ quantiles $b_1^*(\alpha/2)$ and $b_1^*(1 - \alpha/2)$
- Percentiles from b_1 :

$$b_1^*(\alpha/2) \leq \beta_1 \leq b_1^*(1 - \alpha/2)$$

- **reflection method**: require a larger number of bootstrap samples

$$2b_1 - b_1^*(\alpha/2) \leq \beta_1 \leq 2b_1 - b_1^*(\alpha/2)$$